

35. The Desert Ant

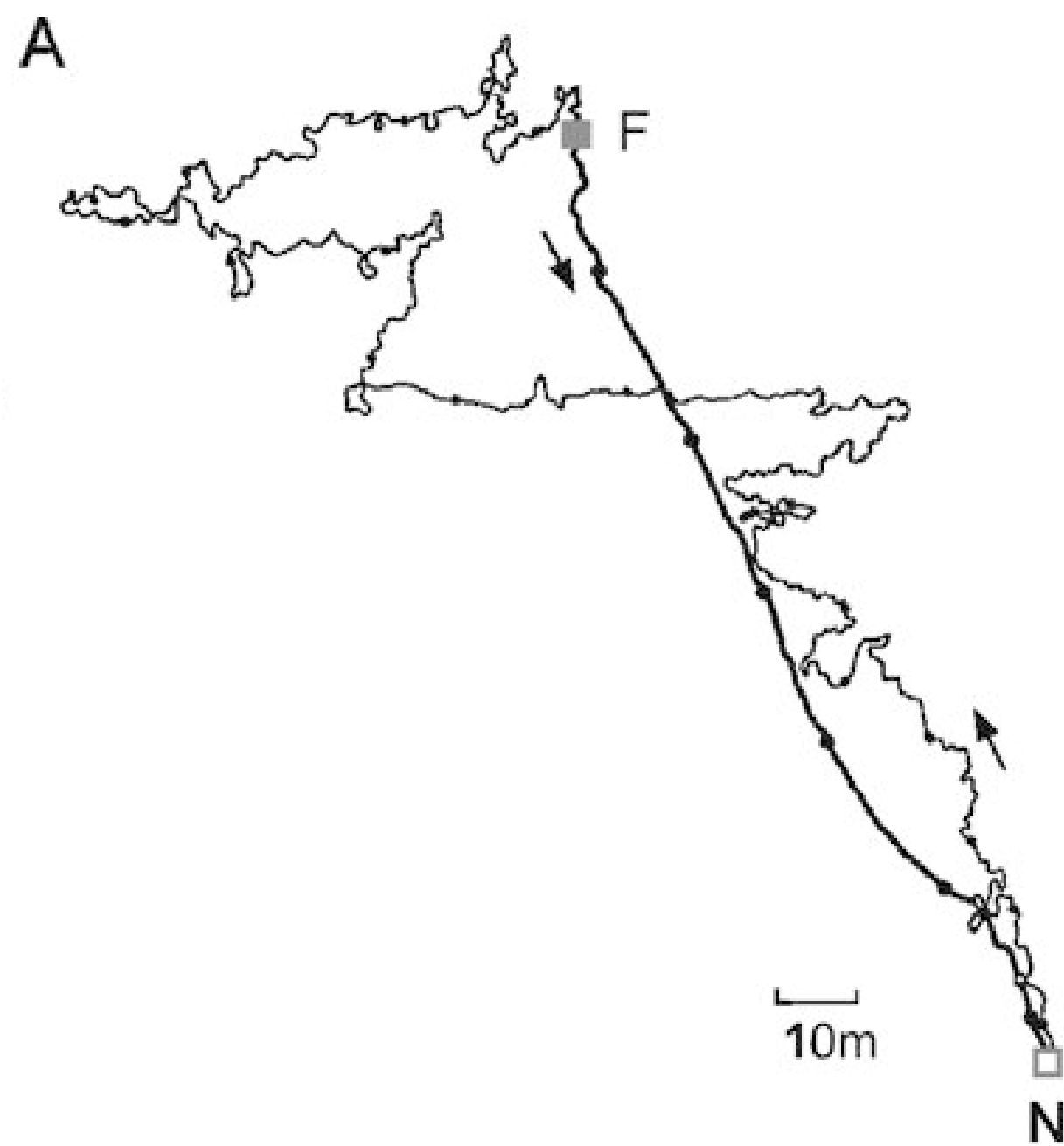
The Ant

Saharan desert ant
Cataglyphis

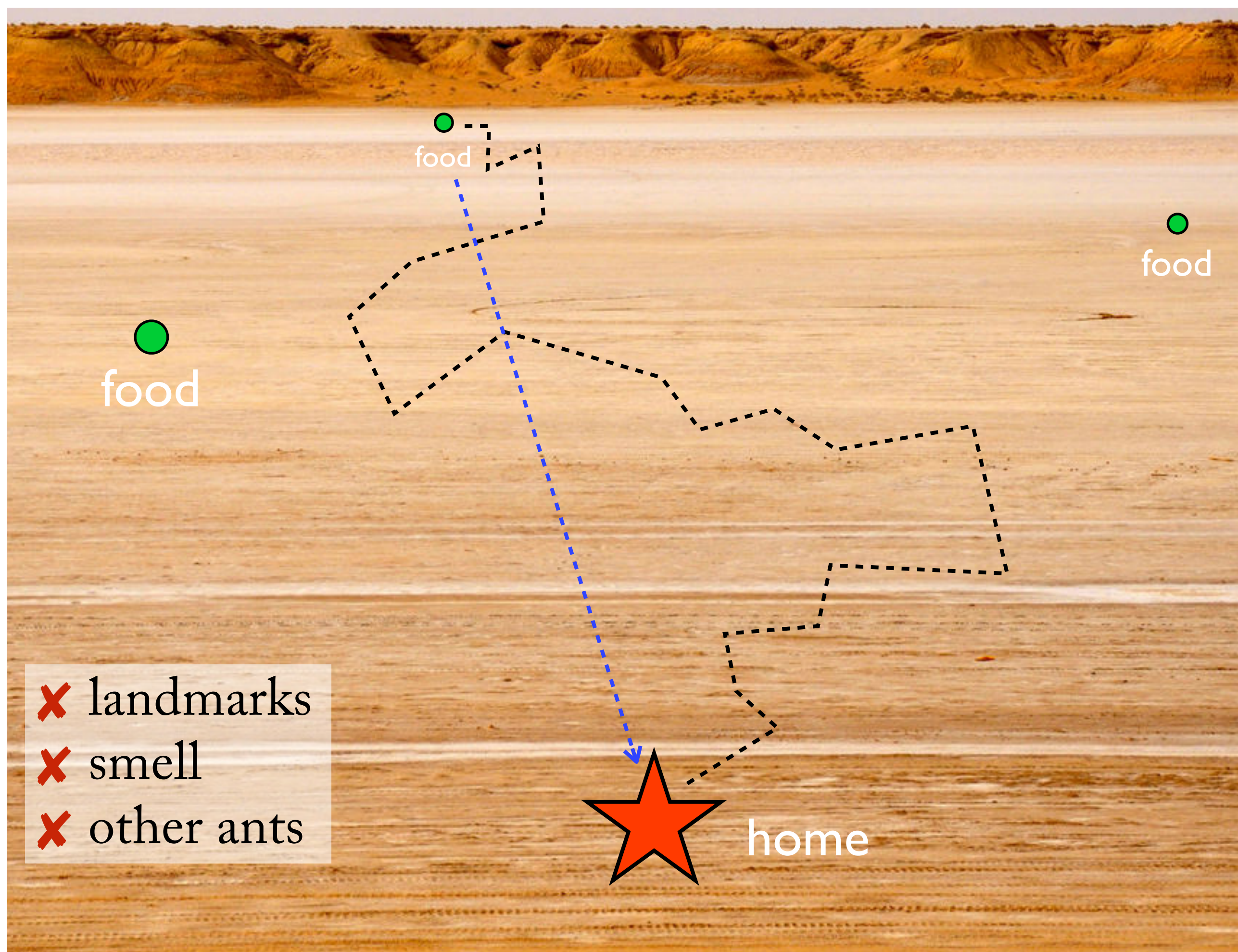


250,000 neurons

The Journey



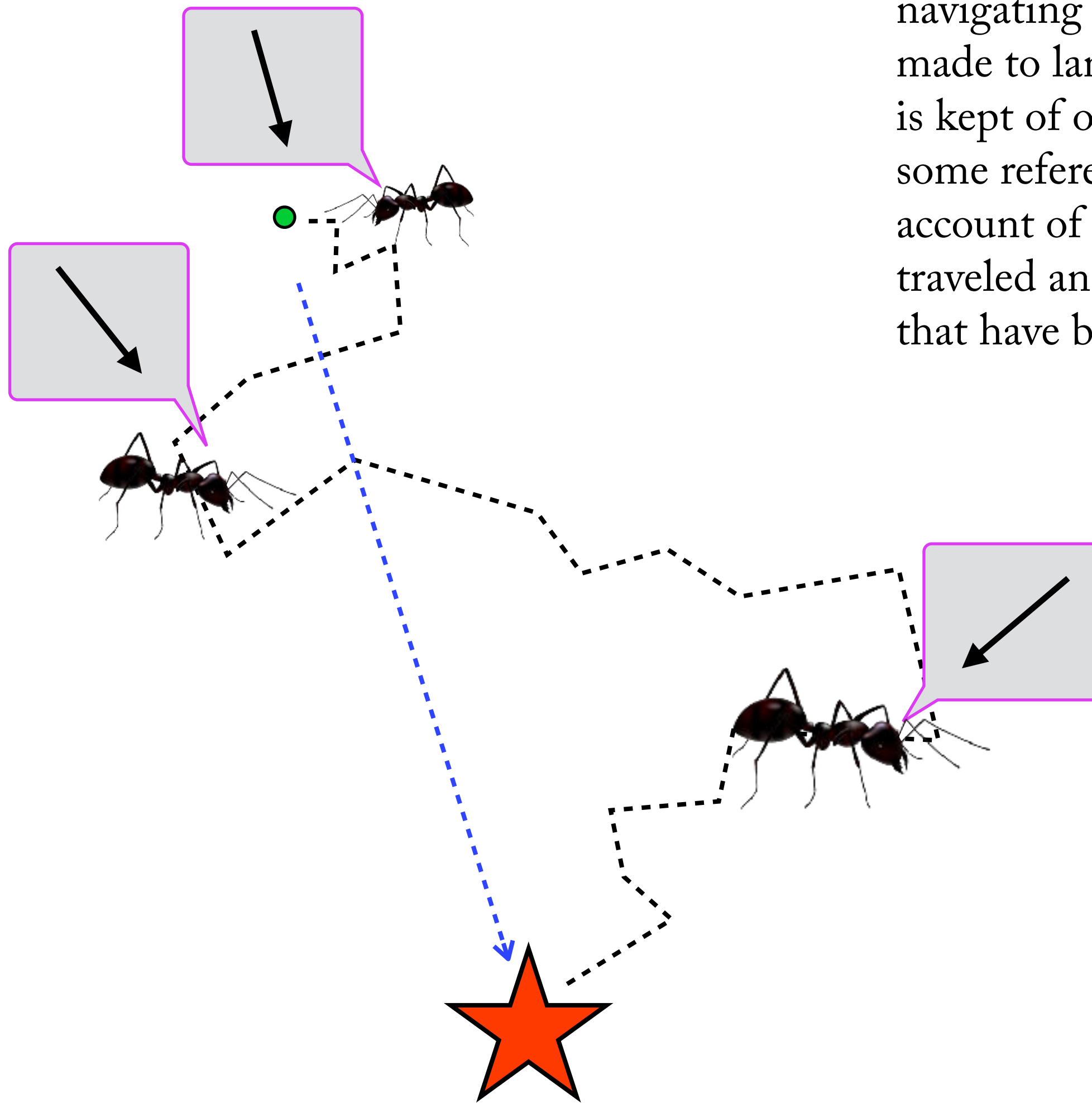
“Scientists observed that instead of retracing their winding route back home, they head straight back to the nest. How does the ant keep track of its current location relative to home? Path integration is the name given to the process thought to be used by animals (human and non-human alike) for what some might call dead reckoning.” — SA



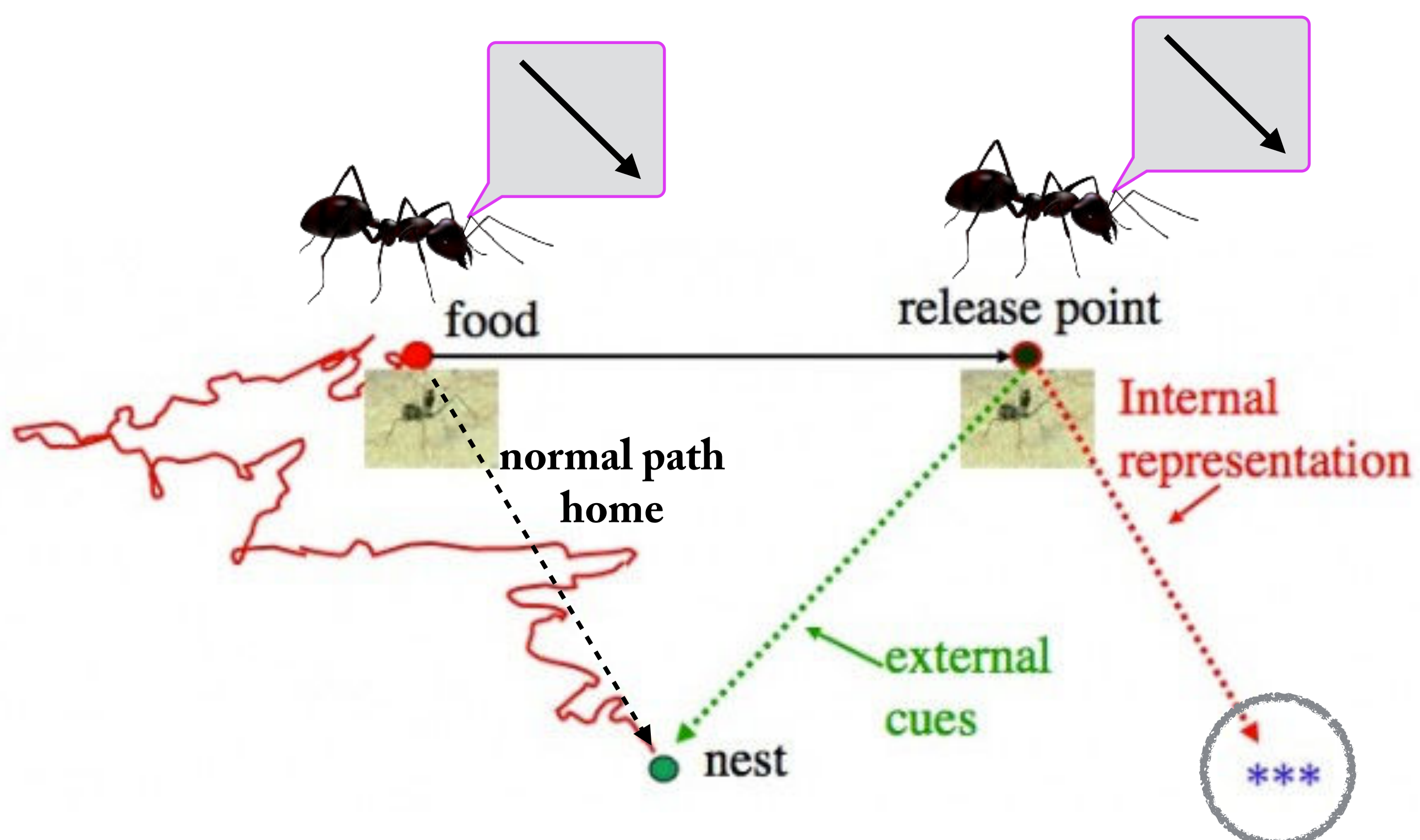
Ant = ~ 0.5 cm
Travel = >150 m
= over 20,000x
their length

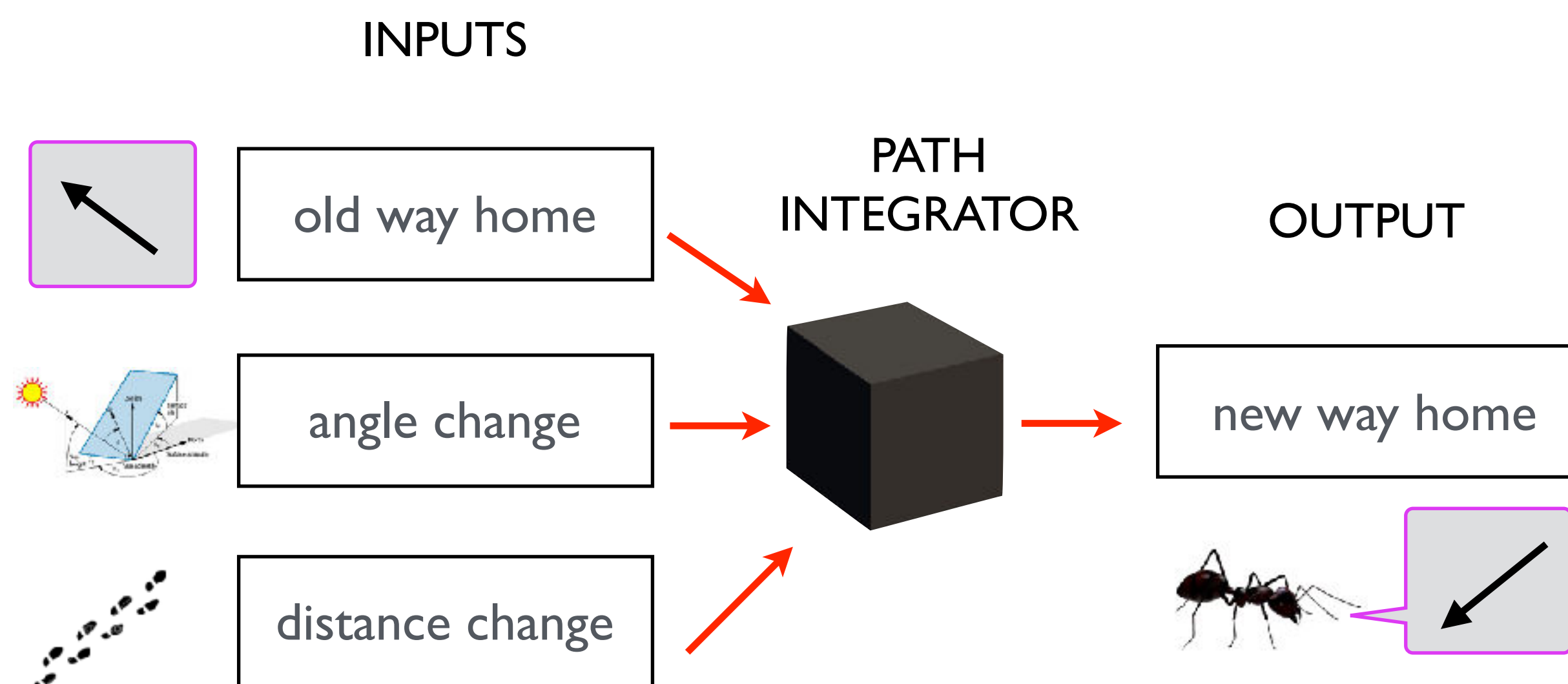
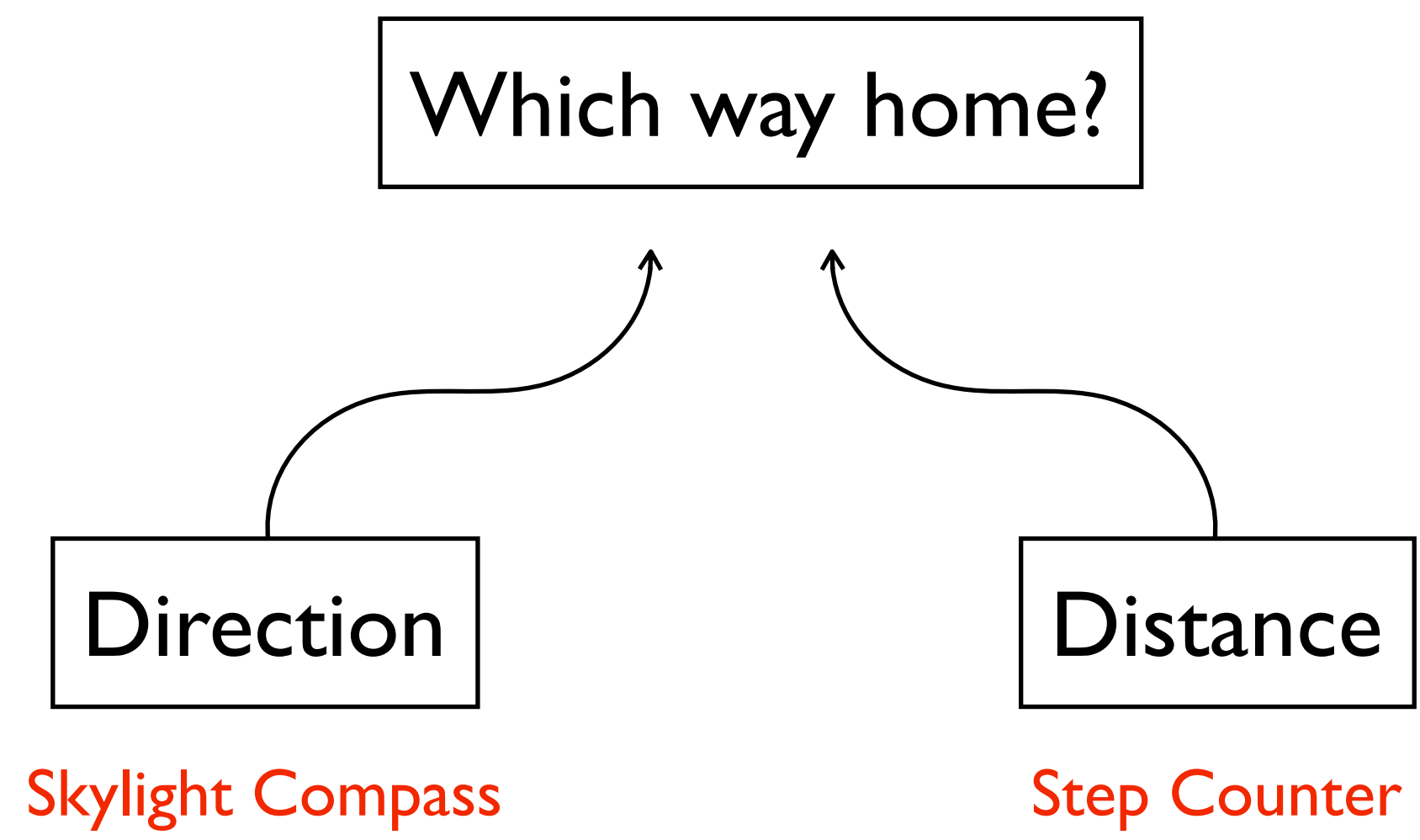
The Representational Hypothesis

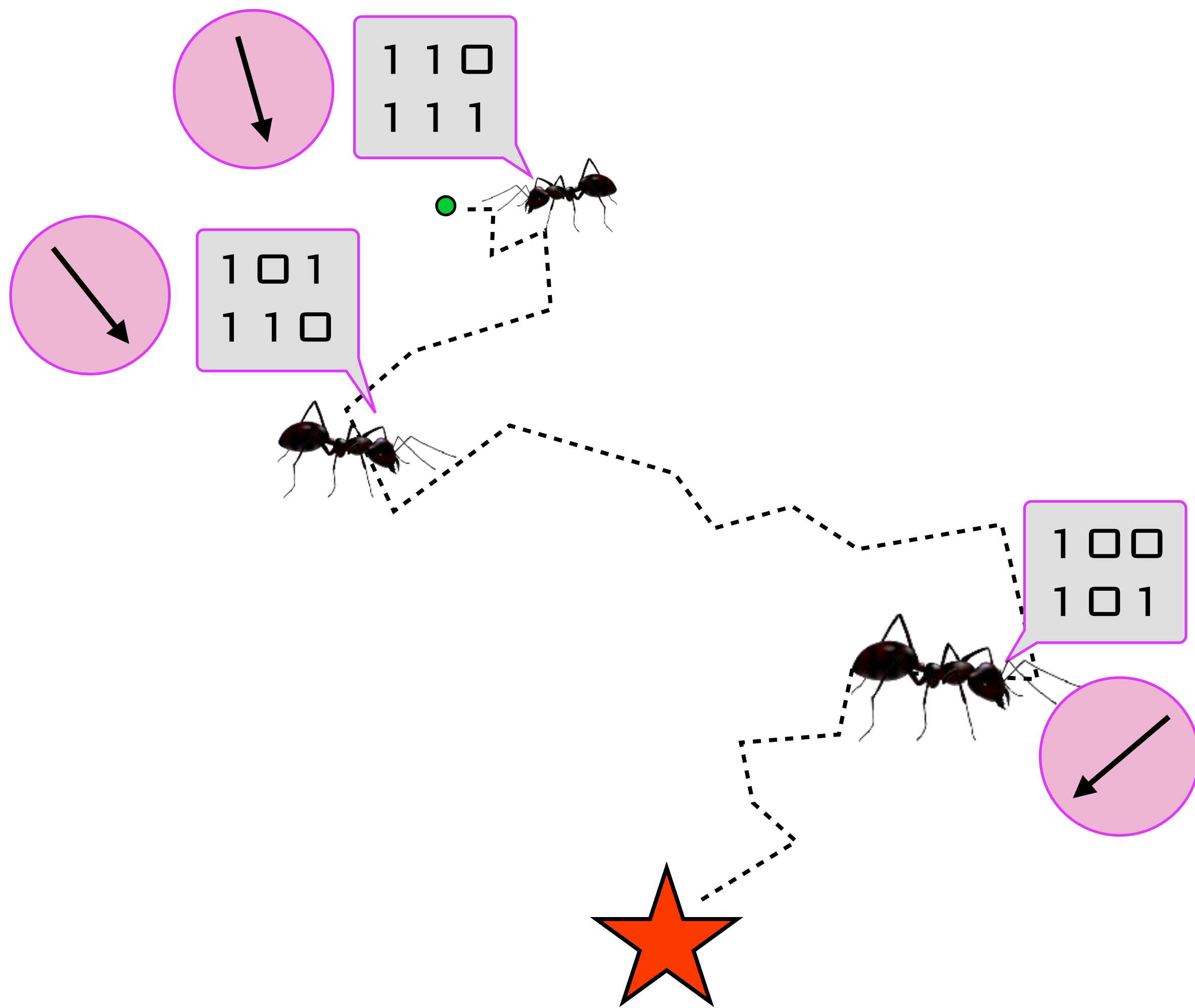
“**Dead reckoning**... is the method of navigating where no reference is made to landmarks. Instead, a record is kept of one's position in respect to some reference point by taking account of the distance that has been traveled and the changes in direction that have been made.” Pearce 2008



Testing the Representational Hypothesis







The Desert Ant

First, search until you find food (are placed over it).
Second, return to where you started as efficiently as possible.

